Essays

|!|BCEEC0160000001|!|

What is eye accommodation? Using daily examples to illustrate how eye accommodation works. (12 marks)

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- Eye accommodation refers to the ability of the eye to focus on objects located at different distances. (1)
- It is brought about by the action of the circular ciliary muscles, which contract or relax to change the tension in the suspensory ligaments. (1)
- This changes the shape / curvature of the lens so that light rays can be focused sharply onto the retina. (1)
- When we are reading a book (or other examples), our eyes are focusing on a near object. (1)
- The circular ciliary muscles contract, reducing the tension in the suspensory ligaments. (1)
- Since the lens is elastic, it becomes thicker (more convex) to refract light rays more. (1)
- When we are looking at the blackboard (or other examples), our eyes are focusing on a distant object. (1)
- The circular ciliary muscles relax, increasing the tension in the suspensory ligaments. (1)
- The lens is pulled to become thinner (less convex) to refract light rays less. (1)

Communication (Max. 3)

(12 marks)

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Skills assessed	Marks
Understanding basic concepts	7
Applying concepts	2
Communication	3

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|!|BCEEC0160000002|!|

When a fire alarm rings, we are able to hear the bell. Describe how the ears and the nervous system are involved in this hearing process. (12 marks)

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- The fire alarm sets off sound waves which are collected by the pinna. (1)
- Sound waves are directed along the auditory canal to the eardrum (1)
- The eardrum converts the sound waves into vibrations. (1)
- The ear bones in the middle ear amplify and transmit vibrations from the eardrum to the oval window. (1)
- The oval window vibrates and causes wave movements of the perilymph in the upper canal of the cochlea. (1)
- The wave movements are transmitted to the endolymph in the central canal of the cochlea. (1)
- Hairs of sensory hair cells in the central canal are bent and the cells generate nerve impulses. (1)
- Nerve impulses are transmitted along the auditory nerve (1)
- to the auditory centre of the brain to produce the sensation of hearing. (1)

Communication (Max. 3)

(12 marks)

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Skills assessed	Marks
Understanding basic concepts	9
Communication	3

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